

Cheat Sheet :: VEGAN



What is VEGAN?

The **vegan** package provides tools for descriptive community ecology. It has basic functions of **community ordination**, **diversity analysis** and **dissimilarity analysis**. Most of its multivariate tools can be used for other data types as well.

Examples using : **data(dune)**

Unconstrained Ordination

metaMDS(data,...) **Nonmetric Multidimensional Scaling**

All ordination results can be displayed with

plot(data, type = "")

type = "p" results with points of black circles to indicate sites and red pluses to show species

type = "t" results with text

ordihull() adds convex hulls

ordiellipse() adds ellipses of standard deviation, standard error or confidence areas

ordispider() draws items to their center



Constrained Ordination

cca(formula, data, ...) **Constrained Correspondence Analysis**
Displays only the variation that can be explained by used constraints

rda(formula, data, scale=FALSE, ...) **Redundancy Analysis**

capscale(formula, data, distance = "", ...) **Distance based Redundancy Analysis**

formula() Model formula must be either community data matrix or dissimilarity matrix

OR

distance = "name of dissimilarity index" if formula is not specified

Analysis of constraints

anova.cca(object, permutations = "", ...) **Permutation Test for CCA & RDA to assess the significance of constraints**

object specifies one or several result objects from cca, rda, or capscale

permutations = control values, or permutation index

Diversity Analysis of Eco Communities

diversity(data, index = "", MARGIN = 1, base = exp(1), ...) **Shannon, Simpson, and Fisher diversity indices and species richness.**

renyi(data, scale = c(), raw = FALSE, ...) **Rényi Diversity index**



scale = c() Scales of Rényi diversity

raw = FALSE summary stats of permutations

raw = TRUE individual permutations

rarefy(data, sample, ...) **Rarefied Species Richness results in expected species richness of subsample**

Taxonomic Diversity

taxondive(data, distance, match.force = FALSE) **Taxonomic diversity indices**

taxa2dist(data, varstep = FALSE, check = TRUE, ...) **Converts class tables to taxonomic distances**

Ranked Abundance Distribution

radfit(data, ...) **Fits the most popular model to data using maximum likelihood estimation**

rad.null(data, family = poisson) **Fits broken stick model to expected abundance of species**

type = "b" Plots both observed points and fitted lines

family = Error distribution; poisson default is used for counts, gaussian may be appropriate for abundance



Beta Diversity

betadiver(data, method = NA, ...) **Estimates beta diversity**

method = "" can specify which beta index to use (24 options)

betadiver(help=TRUE) list all 24 indices available

Analysis of Diversity in Groups

anosim(data, grouping, permutations = "", distance = "", ...) **Analysis of similarities between two or more groups**



Dissimilarity Analysis

vegdist(data, method = "", na.rm = FALSE, ...) **Dissimilarity indices**

method = "dissimilarity index"

> "manhattan", "euclidean", "canberra", "clark", "bray", "kulczynski", "jaccard", "gower", "altGower", "morisita", "horn", "mountford", "raup", "binomial", "chao", "cao" or "mahalanobis".

Other Fun Features

vegemit(data, use, scale, sp.ind = "", site.ind = "", select, ...) **Creates a compact ordered community tree in text format**



tabasco(data, use, sp.ind = "", site.ind = "", select, Rowv = TRUE, Colv = TRUE, scale, col = heat.colors(12), ...) **Creates a community table using heat map, abundances are coded by color**

use is either a vector or object

sp.ind / site.ind species and site indices

select a subset of plots

Rowv / Colv = reorder rows and columns, if TRUE it is ordered by correspondence analysis



beals(data, species = NA, reference = data, include = TRUE)

Beals Smoothing and Degree of Absence Analysis determines probability of a species occurring in a site based on joint occurrences with other species

species = NA will compute for all species, or can specify single

reference = data to be used to compare for joint analysis

include = TRUE to include target species in computations

****VEGAN** uses quantitative data but setting **binary = TRUE** will make data presence/absence**